



Assignment 7 Diya Jain U81673988

$$1. A = \begin{bmatrix} 4 & -1 & 6 \\ 2 & 1 & 6 \\ 2 & -1 & 8 \end{bmatrix} \quad (A - \lambda I)$$

$$\begin{bmatrix} 4-\lambda & -1 & 6 \\ 2 & 1-\lambda & 6 \\ 2 & -1 & 8-\lambda \end{bmatrix}$$

Finding determinant.

$$C_1 = C_1 - (\lambda - 4)C_2$$

$$\begin{vmatrix} 0 & -1 & 6 \\ (\lambda-3)(\lambda-2) & 1-\lambda & 6 \\ (\lambda-2) & -1 & 8-\lambda \end{vmatrix}$$

$$C_3 = C_3 + 6C_2$$

$$\begin{vmatrix} 0 & -1 & 0 \\ (\lambda-3)(\lambda-2) & 1-\lambda & 12-6\lambda \\ \lambda-2 & -1 & 2-\lambda \end{vmatrix}$$

Expanding along row 1:

$$\begin{vmatrix} (\lambda-3)(\lambda-2) & 12-6\lambda \\ \lambda-2 & 2-\lambda \end{vmatrix}$$

$$= ((\lambda-3)(\lambda-2)) \cdot (2-\lambda) - (12-6\lambda) \cdot (\lambda-2)$$

$$= -(\lambda-9)(\lambda-2)^2$$



The roots are $\lambda_1 = 9$, $\lambda_2 = 2$, $\lambda_3 = 2$

Then eigenvectors;

a. $\lambda = 9$

$$\begin{bmatrix} 4-\lambda & -1 & 6 \\ 2 & 1-\lambda & 6 \\ 2 & -1 & 8-\lambda \end{bmatrix} = \begin{bmatrix} -5 & -1 & 6 \\ 2 & -8 & 6 \\ 2 & -1 & -1 \end{bmatrix}$$

Null space of this matrix is $\left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}$

b. $\lambda = 2$

$$\begin{bmatrix} 4-\lambda & -1 & 6 \\ 2 & 1-\lambda & 6 \\ 2 & -1 & 8-\lambda \end{bmatrix} = \begin{bmatrix} 2 & -1 & 6 \\ 2 & -1 & 6 \\ 2 & -1 & 6 \end{bmatrix}$$

Null space of this matrix is $\left\{ \begin{bmatrix} 1/2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix} \right\}$

$$S = \begin{bmatrix} 1 & 1/2 & -3 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \quad D = \begin{bmatrix} 9 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$$A = S D S^{-1} = \begin{bmatrix} 1 & 1/2 & -3 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 9 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$



2. a. $A = \begin{bmatrix} 2 & -3 \\ 4 & -5 \end{bmatrix}$

$$\begin{bmatrix} 2-\lambda & -3 \\ 4 & -\lambda-5 \end{bmatrix}$$

$$\det(A) = (2-\lambda)(-\lambda-5) + 12$$

$$= -2\lambda - 10 + \lambda^2 + 5\lambda + 12$$

$$= \lambda^2 + 3\lambda + 2$$

$$= \lambda^2 + 2\lambda + \lambda + 2$$

$$= \lambda(\lambda+2) + 1(\lambda+2)$$

$$= (\lambda+2)(\lambda+1)$$

roots are $\lambda_1 = -2$, $\lambda_2 = -1$.

Eigenvectors

a. $\lambda = -1$

$$\begin{bmatrix} 2-\lambda & -3 \\ 4 & -\lambda-5 \end{bmatrix} = \begin{bmatrix} 3 & -3 \\ 4 & -4 \end{bmatrix}$$

$$= \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}$$

b. $\lambda = -2$

$$\begin{bmatrix} 2-\lambda & -3 \\ 4 & -\lambda-5 \end{bmatrix} = \begin{bmatrix} 4 & -3 \\ 4 & -3 \end{bmatrix} = \left\{ \begin{bmatrix} 3|4 \\ 1 \end{bmatrix} \right\}$$



b. $A = PDP^{-1}$

$$P = \begin{bmatrix} 1 & 3/4 \\ 1 & 1 \end{bmatrix}$$

$$D = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix}$$

$$A^2 = (PDP^{-1})(PDP^{-1}) \\ = PD^2P^{-1}$$

$$A^6 = PD^6P^{-1}$$

$$= \begin{bmatrix} 1 & 3/4 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix}^6 \begin{bmatrix} 4 & -3 \\ 4 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 3/4 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 64 \end{bmatrix} \begin{bmatrix} 4 & -3 \\ -4 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 3/4 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 4 & -3 \\ -256 & 256 \end{bmatrix}$$

$$= \begin{bmatrix} -188 & 189 \\ -262 & 253 \end{bmatrix}$$



3. a. $A = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 \end{bmatrix}$

$$\begin{bmatrix} -\lambda-1 & 0 & 0 & 0 \\ 0 & -\lambda & 0 & 0 \\ 0 & 0 & -\lambda & 0 \\ 0 & 0 & 1 & -\lambda-1 \end{bmatrix}$$

$$\det = \lambda^2(\lambda+1)^2$$

roots are $\lambda_1 = -1, \lambda_2 = -1, \lambda_3 = 0, \lambda_4 = 0$

• $\lambda = -1$

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

• $\lambda = 0$

$$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}$$

$$P = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \end{bmatrix}, D = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$



b. $A = PDP^{-1}$

$$A^2 = PD^2P^{-1}$$

$$A^{2022} = PD^{2022}P^{-1}$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 2 & 1 & 2 \\ 0 & 2 & 8 & 0 \\ 0 & 1 & 1 & 6 \end{bmatrix}$$